Traffic Collision Data Analysis

Objective

In this case study, you will be working on California Traffic Collision Data Analysis using Apache Spark, a powerful distributed computing framework designed for big data processing. This assignment aims to provide hands-on experience in analyzing large-scale traffic collision datasets using PySpark and AWS services. You will apply data analytics techniques to clean, transform, and explore crash data, drawing meaningful insights to support traffic safety and urban planning. Beyond understanding how big data tools optimize performance on a single machine and across clusters, you will develop a structured approach to analyzing crash trends, identifying high-risk locations, and evaluating contributing factors to traffic incidents. Additionally, you will utilize AWS S3 to store the processed data efficiently after the ETL process, enabling scalable storage and easy retrieval for further analysis.

Business Value

Traffic collisions pose significant risks to public safety, requiring continuous monitoring and analysis to enhance road safety measures. Government agencies, city planners, and policymakers must leverage data-driven insights to improve infrastructure, optimize traffic management, and implement preventive measures.

In this assignment, you will analyze California traffic collision data to uncover patterns related to accident severity, location-based risks, and key contributing factors. With Apache Spark's ability to handle large datasets efficiently and AWS S3's scalable storage, transportation authorities can process vast amounts of crash data in real time, enabling faster and more informed decision-making.

As an analyst examining traffic safety trends, your task is to analyze historical crash data to derive actionable insights that can drive policy improvements and safety interventions. Your analysis will help identify high-risk areas, categorize accidents by severity and contributing factors, and store the processed data in an AWS S3 bucket for scalable and long-term storage.

By leveraging big data analytics and cloud-based storage, urban planners and traffic authorities can enhance road safety strategies, reduce accident rates, and improve public transportation planning.

Dataset Overview

The dataset used in this analysis consists of California traffic collision data obtained from the Statewide Integrated Traffic Records System (SWITRS). It includes detailed records of traffic incidents across California, covering various attributes such as location, severity, involved parties, and contributing factors. The dataset has been preprocessed and transformed using PySpark to facilitate large-scale analysis. By leveraging Apache Spark, we ensure efficient data handling, enabling deeper insights into traffic patterns, accident trends, and potential safety improvements.

The dataset is a .sqlite file contains detailed information about traffic collisions across California and is structured into four primary tables:

- collisions table contains information about the collision, where it happened, what vehicles were involved.
- parties table contains information about the groups people involved in the collision including age, sex, and sobriety.

- victims table contains information about the injuries of specific people involved in the collision.
- locations table contains information about the geographical location and details of road intersections.

Assignment Tasks

1. Data Preparation

The dataset consists of structured tables containing traffic collision data. Before conducting any analysis, it is essential to ensure that the data is properly formatted and structured for efficient processing. Check for data consistency and ensure all columns are correctly formatted. Apply sampling techniques if needed to extract a representative subset for analysis. Structure and prepare the data for further processing and analysis.

Spark session is created with the name "Traffic Collision Analysis". All 4 datasets were read into separate spark dataframes.

2. Data Cleaning

2.1 Handling Missing Values

There are some columns with significant number of missing values except case_df. To handle missing values, the percentage of missing values was calculated for each column. A threshold of 10% was set. Any column with a percentage of missing values above this threshold will be considered sparse and dropped from the DataFrame. This helps to remove columns that contribute little to the analysis due to excessive missing data.

There are some date and time columns with string datatypes. These can be converted to datadatetype and timestamptype data types respectively. To handle the missing values in all the spark dataframes, if the number of nulls is less than 1%, then drop the rows. For nulls in categorical columns - replace nulls with the most frequent value. For nulls in numeric columns - replace nulls with median value. For nulls in date columns - drop the corresponding rows.

2.2 Fixing Columns

The column names in collisions_df, parties_df, and victims_df have been successfully cleaned and standardized by converting them to snake_case and lowercase to ensure proper naming and formatting. PySpark ensures that there are no syntactically duplicate column names.

2.3 Handling Outliers

From the list of numerical columns from each DataFrame suitable for outlier detection, IQR was calculated, and outliers were detected. There are a significant number of outliers, and dropping this would lead to collisions_df: ~49% removed, parties_df: ~26% removed and victims_df: ~0.7% removed. "Outliers" in this dataset are not errors—they represent real, high-impact events. The data is naturally skewed. Traffic accidents are not evenly distributed. Most are minor, few are extreme. Extreme values are meaningful, not noise. High injury/fatality counts = critical signals. Removing them biases the dataset. Hence, it is decided to not remove outliers.

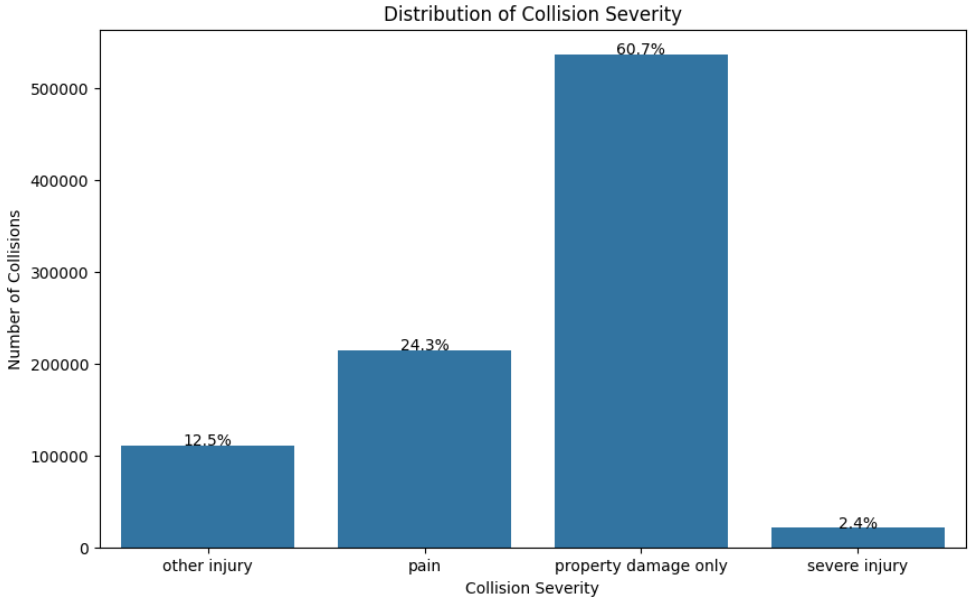
3. Exploratory Data Analysis

Finding Patterns and analyze the dataset and find patterns based on the following points:

3.1 Classify variables into categorical and numerical types.

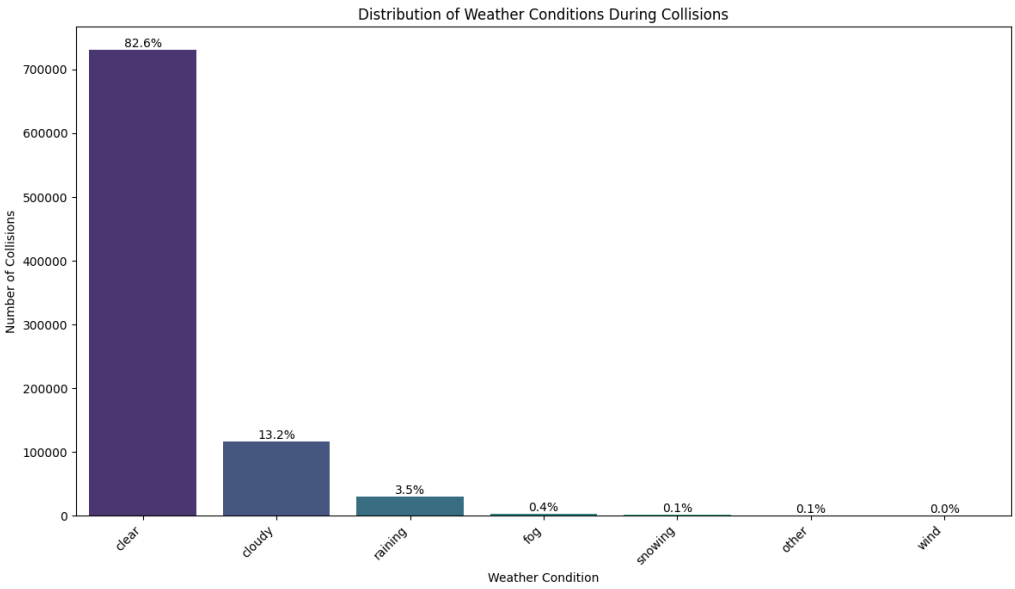
For the dataframes, the categorical and numerical columns were identified. The columns were casted to their respective datatypes and loaded.

3.2 Analyze the distribution of collision severity.



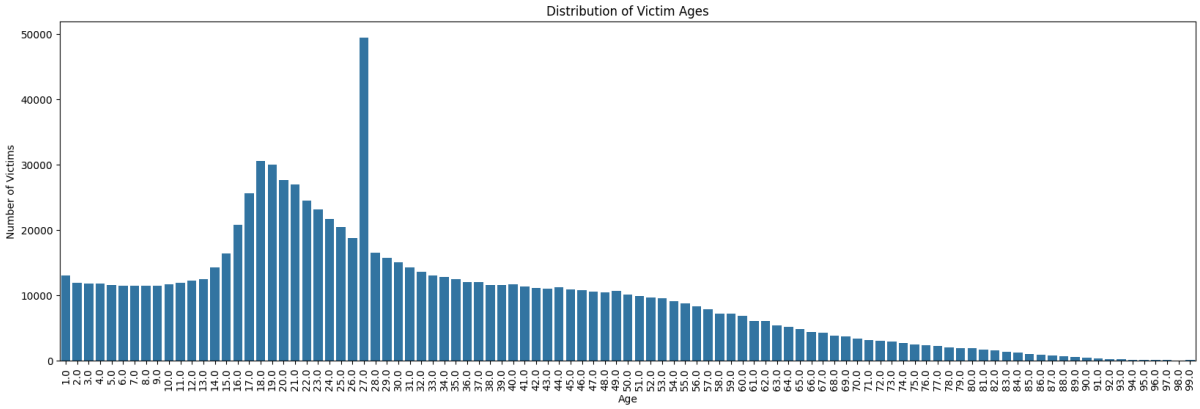
The dataset is heavily skewed toward non-injury collisions. The vast majority of cases involve no physical harm, only vehicle/property damage. But, a significant portion of collisions do involve injuries, not just damage. Severe crashes are low-frequency.

3.3 Examine weather conditions during collisions.



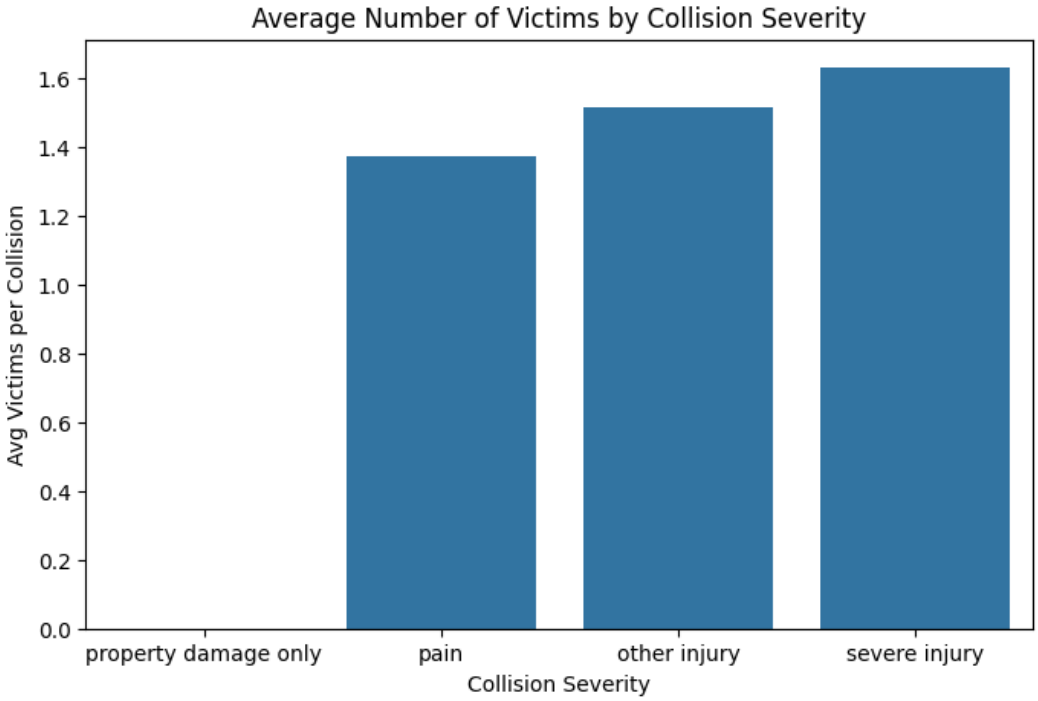
The vast majority of collisions occur in clear weather conditions. Clear weather has the highest number of collisions due to highest exposure, not necessarily highest risk. Collisions during bad weather are relatively infrequent. Since most crashes happen in good weather, it suggests - driver error, distraction, speeding, traffic density, etc are likely more dominant factors than weather alone.

3.4 Analyze the distribution of victim ages.



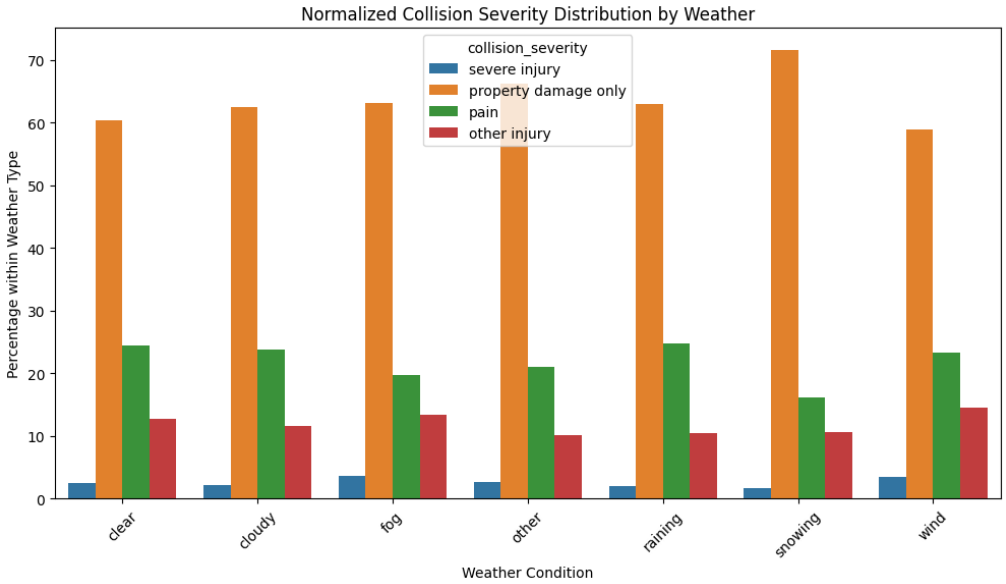
Young adults are the most represented victims in collisions. There is a sudden jump at age ~27 that breaks the smooth pattern. This spike is likely not real-world behavior, but a data quality issue. Gradual decline after age ~30. Collision involvement decreases with age. Older individuals are less frequently involved in collisions. Children (0–15) relatively stable but lower. Likely passengers or pedestrians rather than drivers

3.5 Study the relationship between collision severity and the number of victims.



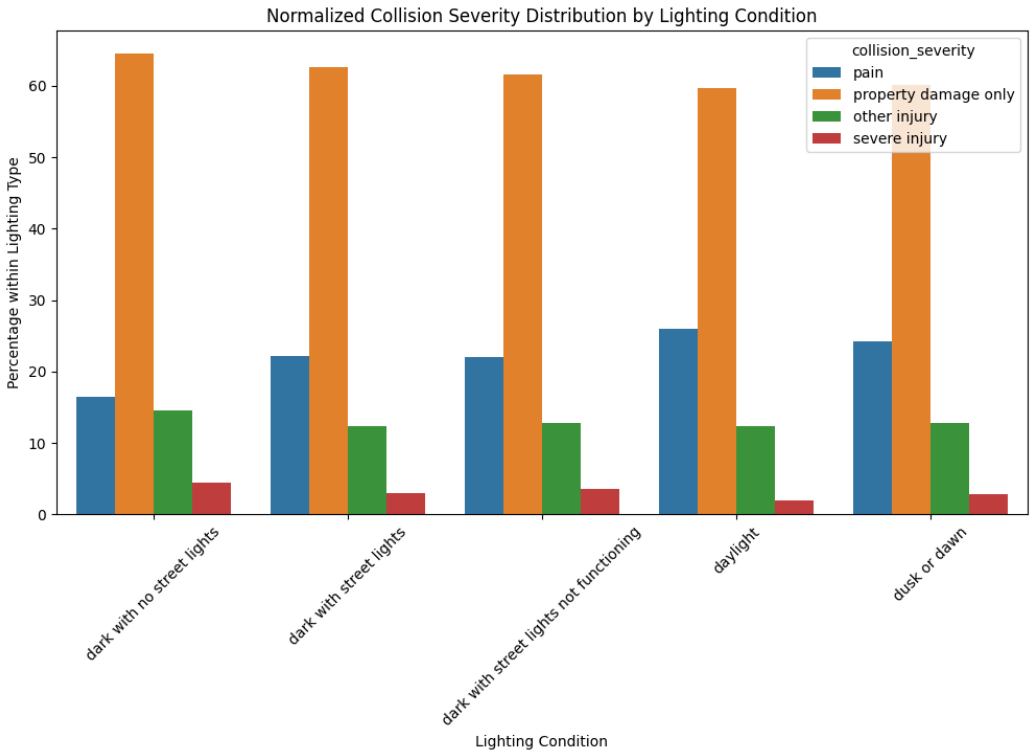
As collision severity increases, the average number of victims per collision also increases. Differences are real but not huge. There is a clear but moderate positive relationship between collision severity and the number of victims, indicating that more severe crashes tend to involve more individuals, though severity is also influenced by factors beyond victim count alone.

3.6 Analyze the correlation between weather conditions and collision severity.



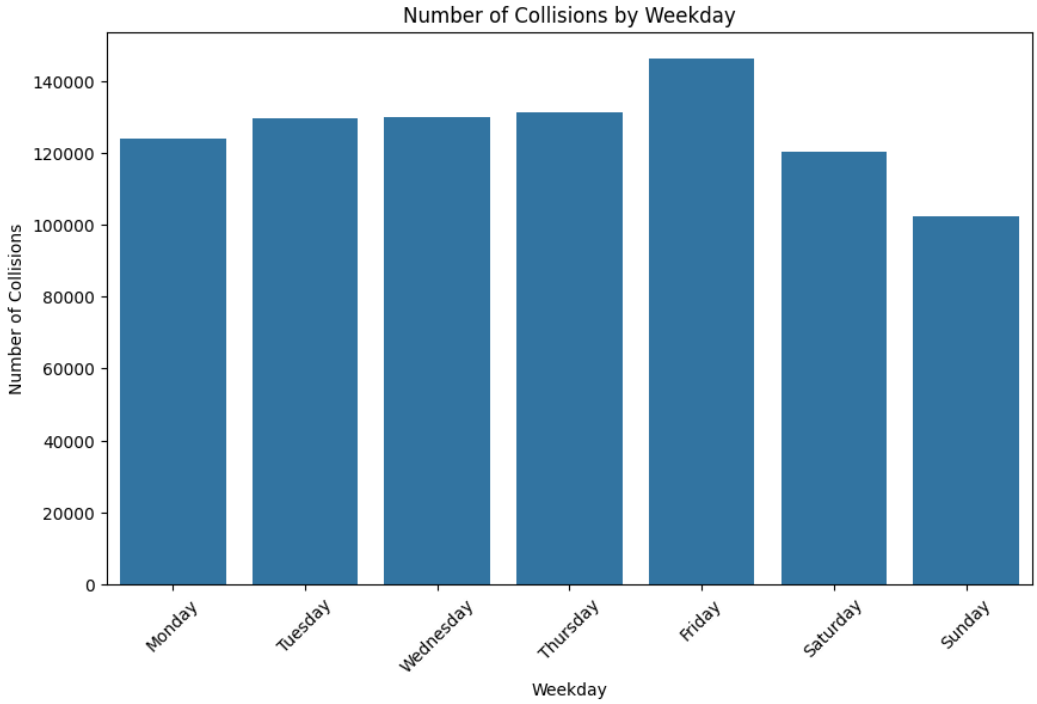
Clear weather has a baseline risk profile, not exceptionally safe. Rainy conditions lead to more injury-prone collisions. Fog is more dangerous per collision. Snow has fewer injuries, but property damage very high. Wind shows higher severe outcomes.

3.7 Visualize the impact of lighting conditions on collision severity.



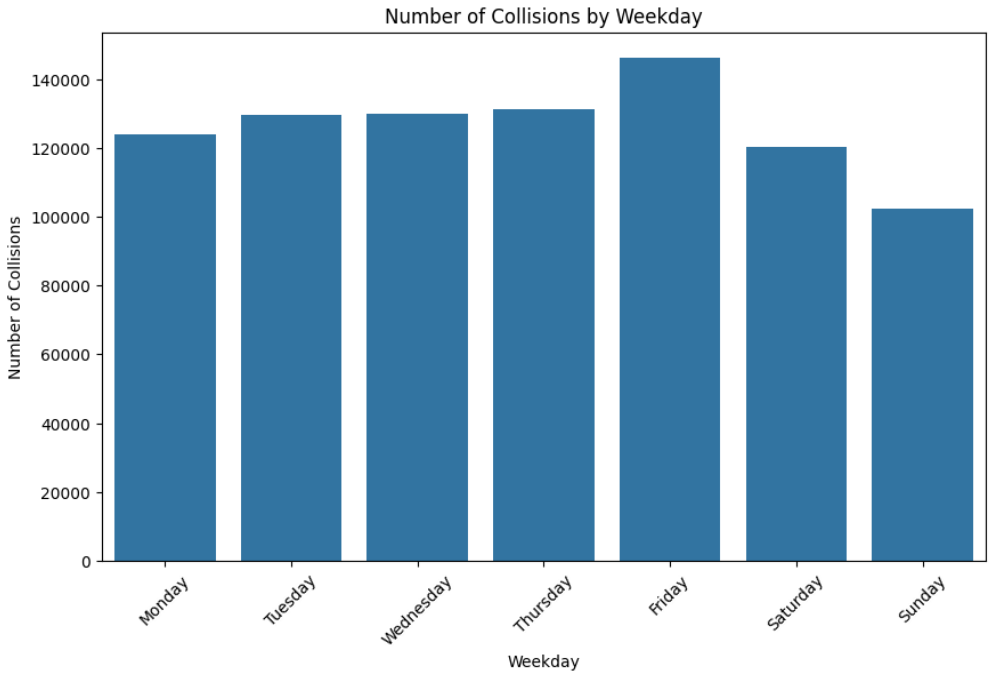
Poor visibility (no lighting) is associated with the highest severity risk. Street lighting mitigates risk, but does not make conditions as safe as daylight. Malfunctioning infrastructure correlates with higher severity outcomes. Malfunctioning infrastructure correlates with higher severity outcomes.

3.8 Extract and analyze weekday-wise collision trends.



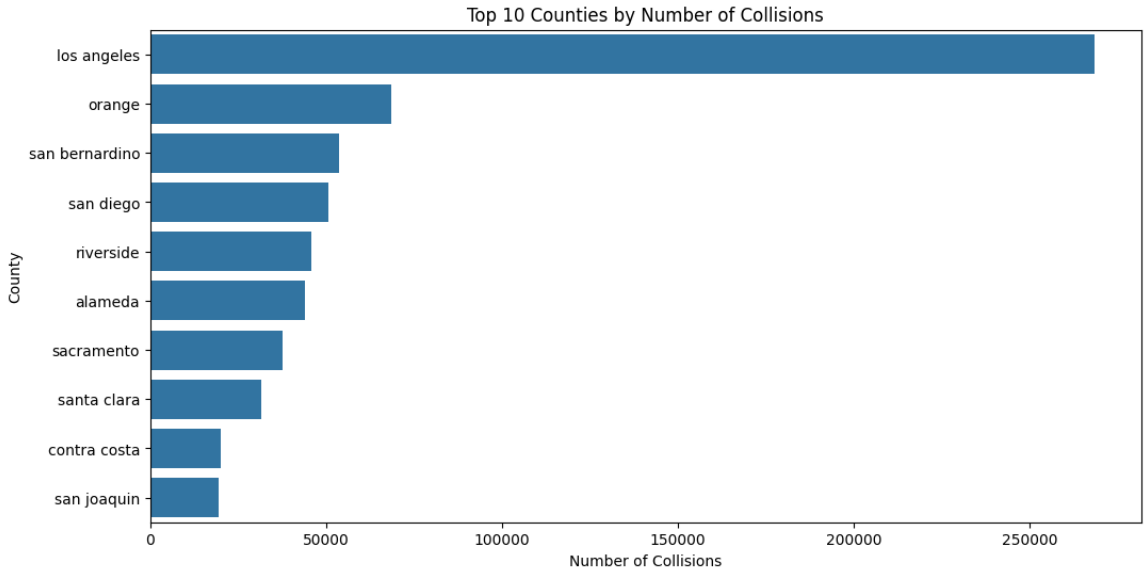
Collision frequency increases throughout the workweek, peaking on Friday due to a combination of high traffic volume and behavioural factors, while weekends—especially Sunday—show significantly fewer collisions, likely reflecting reduced commuting and overall traffic exposure.

3.9 Assess the number of collisions occurring on different days of the week.



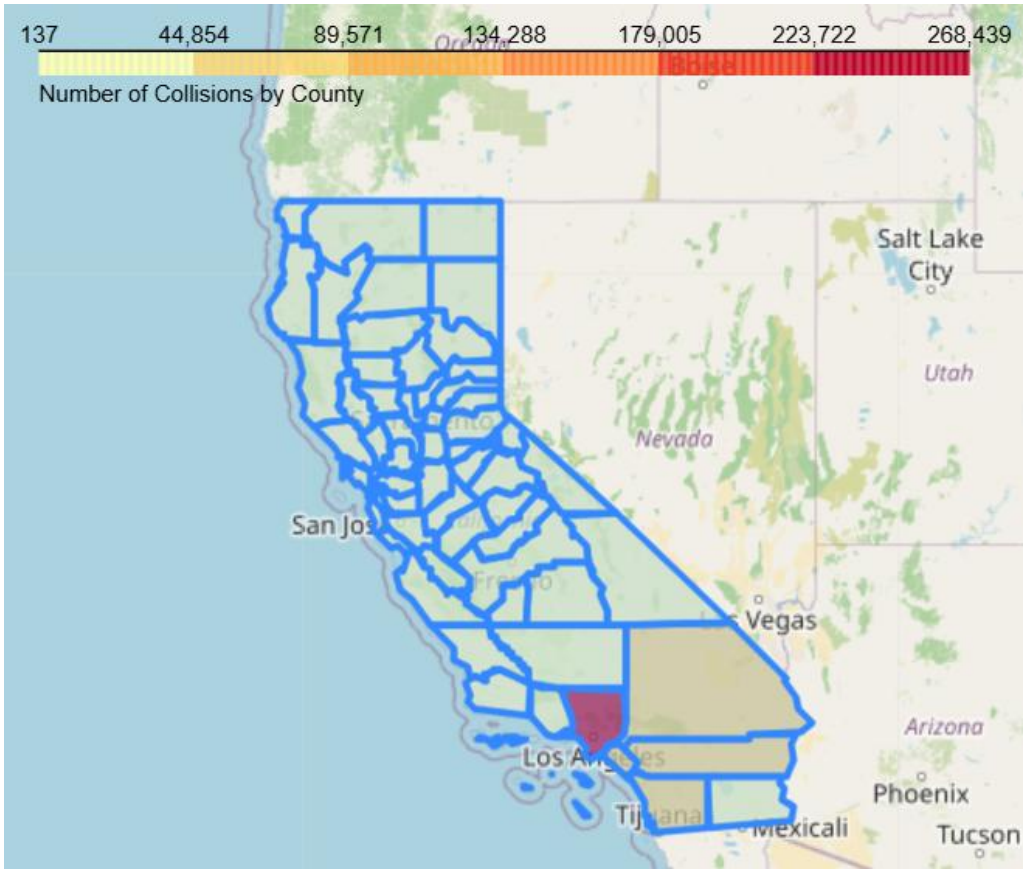
The number of collisions varies systematically across the week, with consistently high levels during weekdays due to commuting activity, peaking on Friday as traffic volume and behavioral factors converge, and declining over the weekend, reaching the lowest levels on Sunday due to reduced traffic exposure.

3.10 Study spatial distribution of collisions by county.



The spatial distribution of collisions is highly uneven, with a strong concentration in major urban counties—particularly Los Angeles—indicating that traffic exposure and population density are the primary drivers of collision frequency.

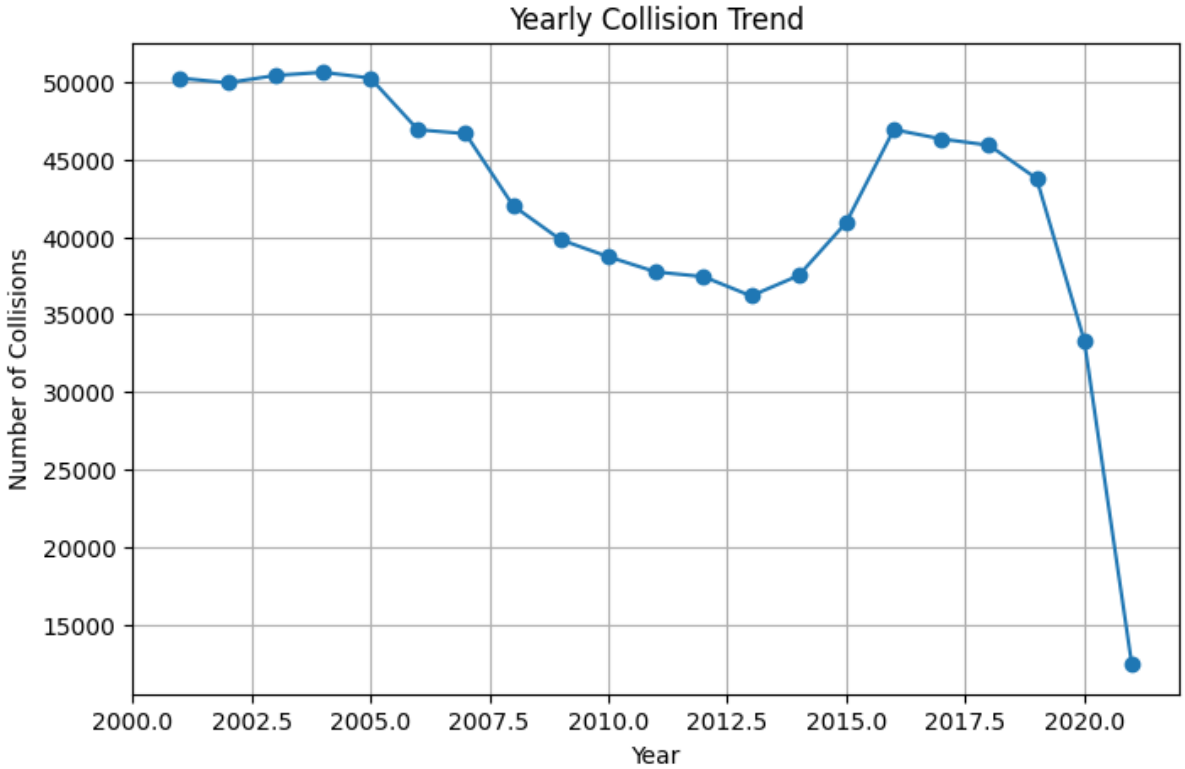
3.11 Generate a scatter plot to analyze collision locations geographically.



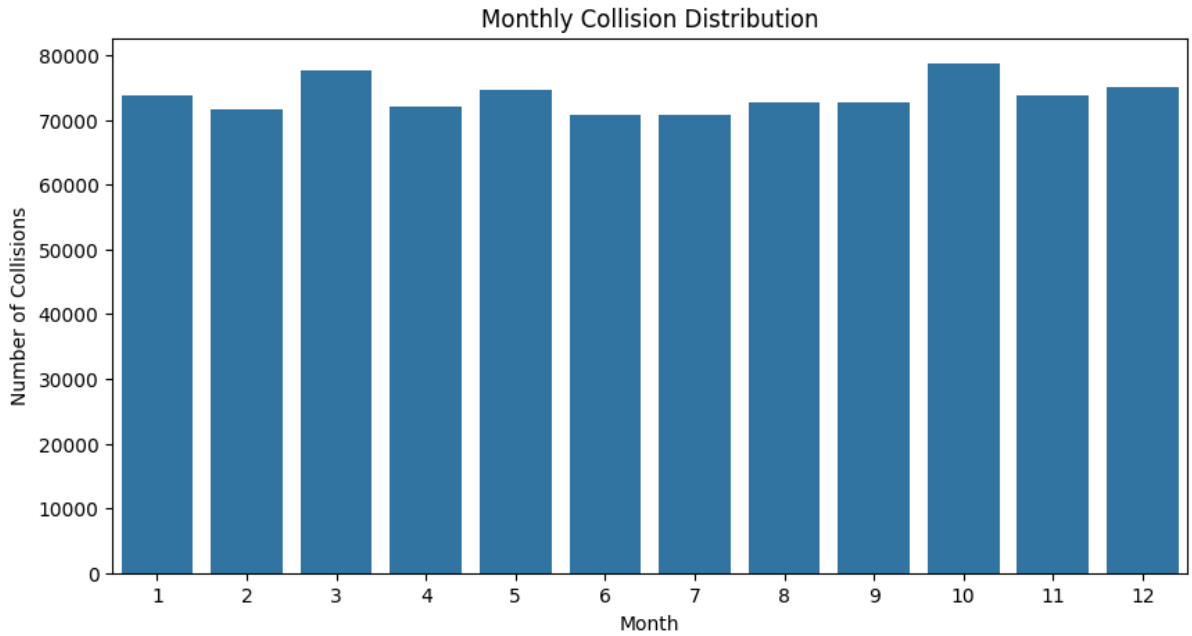
Collision distribution in California is highly concentrated in a few urban counties, with one dominant hotspot. Los Angeles County is the primary collision hotspot in the state. Southern California forms a

major regional hotspot. Northern California contributes significantly but less intensely than Southern California. Central & rural counties have low counts.

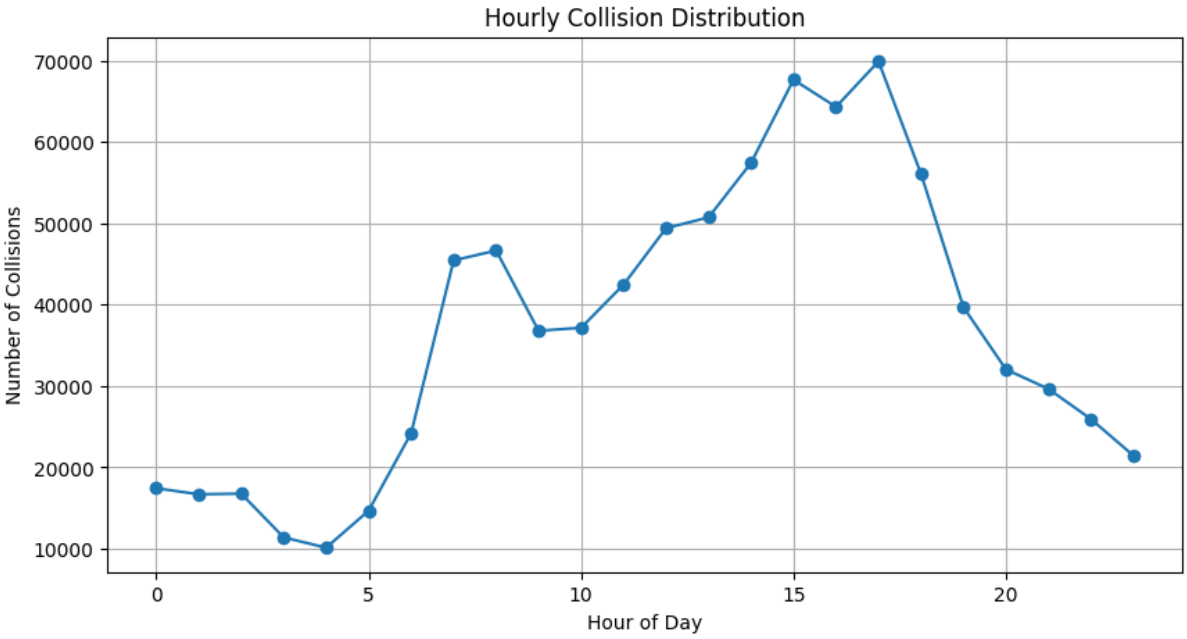
3.12 Extract and analyze collision trends over time, including yearly, monthly, and hourly trends.



Collision counts do not steadily increase—they show phases. Significant decline (2006–2013). Recovery phase (2014–2016). 2020–2021 drop is not likely a real safety improvement. Most likely cause might be COVID-19 lockdowns and reduced mobility. Collision trends are influenced more by societal and mobility changes than by a simple upward growth pattern.



Range for collisions is narrow (~70K–79K). There are no extreme spikes, slight peaks in March and October. Monthly variation is minimal. Collision frequency is relatively stable across months, indicating that seasonality plays a minor role compared to daily traffic patterns.



There is a strong bimodal pattern - two peaks are observed corresponding to the morning and evening commute. Evening peak is dominant, highest collision risk occurs during return commute. Possible reasons for this behaviour could be - Traffic congestion, driver fatigue and rush-hour stress. Collision frequency is primarily driven by commuting behavior, with the highest risk occurring during evening rush hours.

4. ETL Querying

Write PySpark SQL queries for the following:

4.1. Identify the top 5 counties with the highest number of collisions.

The top 5 counties with the highest number of collisions are - los angeles, orange, san Bernardino, san diego, riverside

4.2. Identify the month with the highest number of collisions.

Month with the highest number of collisions – 10 - October

4.3. Determine the most common weather condition during collisions.

The most common weather condition during collisions - clear

4.4. Calculate the percentage of collisions that resulted in fatalities.

The percentage of collisions that resulted in fatalities - 0.7884

4.5. Find the most dangerous time of day for collisions.

Hour 17, ie, 5 PM is the most dangerous time of the day with 69877 collisions.

4.6. Identify the top 5 road surface conditions with the highest collision frequency.

The top road surface conditions with the highest collision frequency are – dry, wet, snowy, slippery

4.7. Analyze lighting conditions that contribute to the highest number of collisions.

The top lighting conditions that contribute to the highest number of collisions – daylight, dark with street lights, dark with no street lights, dusk or dawn, dark with street lights not functioning

5. Conclusion

Provide final insights and recommendations based on the analysis:

5.1 Recommendations to improve road safety by identifying high-risk locations and peak accident times for infrastructure improvements.

The analysis shows that road collisions are heavily concentrated in urban counties with high traffic density, particularly during peak commuting hours, with the highest frequency occurring in the evening rush (4–7 PM) and a secondary peak in the morning. While most collisions happen in daylight and clear weather due to higher exposure, the most dangerous conditions—measured by severity—occur at night, especially in poorly lit areas, where factors like speed, fatigue, and impaired driving increase risk. Seasonal variation is minimal, indicating that daily human activity patterns are the dominant driver of collisions. Based on these insights, improving road safety should focus on targeted infrastructure upgrades such as better street lighting, safer intersection design, and enhanced road visibility in high-risk urban areas, along with time-specific interventions like traffic management during peak hours and stricter enforcement at night. Additionally, protecting vulnerable road users and promoting behaviour change through awareness campaigns can further reduce collision severity and improve overall safety.

5.2 Suggestions to optimize traffic management by analyzing trends in collision severity, weather conditions, and lighting to improve road design and traffic signal timing.

Based on the analysis, traffic management can be significantly optimized by aligning infrastructure design and signal control with observed patterns in collision severity, weather, and lighting conditions. Since collision frequency peaks during high-traffic periods while severity increases under low-visibility conditions (nighttime, poor lighting), adaptive traffic signal timing should prioritize smoother flow during rush hours to reduce congestion-related collisions, while incorporating stricter control measures (longer amber/red phases, reduced speed corridors) during late-night hours to mitigate high-severity incidents. Additionally, integrating weather-responsive traffic systems—such as dynamic speed limits and signal adjustments during adverse conditions (rain, wet roads)—can reduce risk in vulnerable situations. Road design improvements should focus on enhancing visibility through better street lighting, reflective markings, and clearer signage, especially in areas identified with frequent nighttime or low-light collisions. Overall, a data-driven approach that combines temporal patterns, environmental conditions, and severity trends enables more efficient traffic flow while proactively reducing both the frequency and severity of collisions.

5.3 Propose data-driven policy changes to enhance pedestrian and cyclist safety based on collision trends involving vulnerable road users.

To enhance pedestrian and cyclist safety, policies should focus on targeted, data-driven interventions in high-risk urban areas and peak traffic periods. Key measures include expanding protected bike lanes, safer intersection designs with dedicated signals, and adaptive traffic signal timing such as leading pedestrian intervals. Improving visibility through better street lighting and reflective markings is crucial, especially for nighttime safety, while enforcing lower speed limits in pedestrian-heavy zones can reduce injury severity. Additionally, stricter enforcement of traffic laws, combined with public awareness campaigns and continuous monitoring under a Vision Zero framework, can help systematically reduce collisions involving vulnerable road users.

5.4 Identify potential high-risk zones for proactive intervention by examining geographic collision density and historical accident data.

Based on the analysis of geographic collision density and historical accident patterns, high-risk zones can be identified by clustering areas with consistently high collision counts—typically urban counties, major intersections, and busy traffic corridors. These hotspots often coincide with regions of high population density, complex road networks, and heavy commuter traffic. By combining spatial data (e.g., county-level concentration, intersection proximity) with temporal trends (peak hours, recurring accident periods), authorities can pinpoint locations where collisions are both frequent and severe. Proactive interventions should prioritize these zones through targeted measures such as infrastructure redesign (safer intersections, improved signage), enhanced lighting, adaptive traffic signal systems, and increased enforcement. Continuously monitoring these high-risk areas using historical and real-time data enables more effective allocation of resources and supports a preventive, data-driven approach to road safety.

5.5 Assess the impact of environmental factors such as weather, road surface conditions, and lighting on accident frequency and severity.

The analysis indicates that environmental factors influence collision patterns differently for frequency vs severity. Most collisions occur under favourable conditions—clear weather, dry road surfaces, and daylight—primarily due to higher traffic exposure rather than increased risk. However, adverse conditions significantly impact severity rather than volume. Poor lighting conditions (especially dark areas without streetlights) are associated with higher-risk outcomes due to reduced visibility, while wet or compromised road surfaces increase the likelihood of loss of vehicle control. Similarly, although fewer collisions occur during adverse weather, these conditions can elevate crash severity when they do occur. Overall, the findings suggest that environmental factors play a more critical role in amplifying risk and severity rather than driving collision frequency, highlighting the need for targeted interventions such as improved lighting, better road surface maintenance, and adaptive traffic controls during unfavourable conditions.

5.6 Develop predictive models to anticipate collision hotspots and support proactive safety measures.

The analysis can be effectively translated into a predictive, data-driven safety strategy by leveraging historical collision patterns, time-based trends, and environmental factors to identify and anticipate high-risk locations and periods. By developing models that generate risk scores for specific areas and time windows, law enforcement agencies can proactively deploy targeted interventions such as optimized traffic signal timing, focused enforcement, and prioritized infrastructure improvements. This approach enables a shift from reactive incident management to proactive risk mitigation, improving resource allocation, enhancing road safety outcomes, and supporting more informed, strategic decision-making.